BICYCLE COMPANY SALES

Time Intelligence Analysis

Comprehensive Project Documentation

Data Analytics Portfolio Project

Tool: Microsoft Power BI Desktop | DAX Time Intelligence

Data Period: July 2022 – August 2025 | Last Refresh: 26 February 2026

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$77.55M	999.49K	35.13M	0.01
Gross Revenue	Net Profit	YoY Growth	Profit Margin

1. Project Overview & Objectives

This document provides exhaustive technical and analytical documentation for the Bicycle Company Sales — Time Intelligence Analysis project, developed in Microsoft Power BI. The project extends a foundational sales data model with advanced DAX Time Intelligence calculations, enabling comprehensive temporal analysis including year-over-year growth, month-over-month trends, rolling averages, seasonality detection, and YTD performance tracking.

This work is published as a LinkedIn Data Analytics Portfolio project and demonstrates senior-level Power BI competencies, including Calendar Table design, DAX measure engineering, waterfall chart analysis, and executive-level dashboard storytelling.

1.1 Business Questions Answered

- Is revenue increasing year-over-year, and at what rate?
- Which months and quarters generate the highest sales and profit?
- Are there seasonal demand patterns that should inform inventory and staffing decisions?
- Which product categories are growing versus declining over time?
- Which individual products are destroying profit margins?
- How are individual salespeople performing on a year-to-date basis?
- Which regions show the strongest growth trajectory versus the prior year?
- What is the 12-month rolling revenue trend — is the business stable or volatile?

1.2 Key Tools & Technologies

- Microsoft Power BI Desktop — Dashboard design, DAX calculation, data modelling
- Power Query (M Language) — Data transformation, column profiling, type casting
- DAX (Data Analysis Expressions) — All measures, calculated columns, KPIs
- Calendar Table (DAX-generated) — Foundation for all Time Intelligence functions
- Star Schema Model — One fact table (Sales) + four-dimension tables

1.3 Overall Performance Summary (Full Period: Jul 2022 – Aug 2025)

KPI / Metric	Value / Result
Gross Revenue (Total Sales)	$77.55M
Cost of Goods Sold (Total Cost)	$76.55M

Net Profit (Total Profit) Net Profit Margin	$999.49K 1.31% (0.01)
Year-on-Year (YoY) Growth	$35.13M
Month-on-Month (MoM) Growth	$77.55M
Profit Growth (vs Prior Year)	-$240.63K (Negative — Cost Pressure)
Salesperson YTD	$12.65M
Sales Growth %	0.83 (83%)

2. Data Model — Tables, Columns & Quality Profiles

The data model is structured as a Star Schema, which is the industry-standard design for business intelligence and OLAP-style analytical queries. A central fact table (Sales) is surrounded by four dimension tables, with a dedicated Calendar Table enabling all Time Intelligence calculations.

2.1 Salesperson Table

The Salesperson table is a dimension table containing identity, role, and contact information for all 18 sales representatives. It serves as the HR reference dimension for all people-related analytics.

Row Count: 18 | Column Count: 5 | Data Quality: 100% Valid, 0% Error, 0% Empty

Column Statistics (EmployeeKey): Count=18, Min=272, Max=296, Distinct=18, Unique=18

Column	Data Type	Description	Quality	Example
EmployeeKey	Integer	Primary key, unique employee ID. Range 272–296. Joins to Sales table.	100% Valid	272
EmployeeID	Text	HR system reference number. 18 distinct, 18 unique.	100% Valid	502097814
Salesperson	Text	Full name. 18 distinct, 18 unique.	100% Valid	Stephen Jiang
Title	Text	Job title. 5 distinct, 4 unique. Includes North American Sales Manager, Director of Sales, Sales Representative.	100% Valid	Sales Representative

UPN (Email)	Text	Corporate email. 18 distinct, 18 unique. Format: name@adventureworks.com	100% Valid	stephen-jiang@adventureworks.com

2.2 Date Table

The Date table is the original time dimension loaded from source data. It has been supplemented by the purpose-built Calendar Table (see Section 3) for Time Intelligence calculations. The Date table spans 1,461 days.

Row Count: 1,461 | Column Count: 4 | Data Quality: 100% Valid, 0% Error, 0% Empty

Column Statistics (MonthNum): Count=1,000 (profiled sample), Min=1, Max=12, Distinct=12

Column	Data Type	Description	Quality	Profile Stats
Date	Date	Full calendar date. Primary key. 1,000 distinct, 1,000 unique in profile sample.	100% Valid	7/17/2022
Year	Integer	Calendar year. 3 distinct values: 2022, 2023, 2024. 0 unique (repeated).	100% Valid	2022
Month	Text	Month name. 12 distinct, 0 unique (each repeated across years).	100% Valid	July
MonthNum	Integer	Numeric month 1–12 for correct sort ordering. 12 distinct.	100% Valid	7

2.3 Products Table

The Products table is the product master dimension. It contains 397 unique products across 4 top-level categories and 37 subcategories. Standard cost ranges from $0.86 to $2,171.29.

Row Count: 397 | Column Count: 10 | Data Quality: 100% Valid, 0% Error, 0% Empty

Standard Cost Statistics: Min=$0.86, Max=$2,171.29, Average=$436.82, Std Dev=$497.34, Distinct=131, Unique=49

Column	Data Type	Description	Quality	Key Stats

ProductKey	Integer	Primary key. 397 distinct, 397 unique. Joins to Sales[ProductKey].	100%	397 unique
Product	Text	Full product name. 295 distinct, 218 unique.	100%	HL Road Frame - Black, 58
Standard Cost	Decimal	Manufacturing cost per unit. Right-skewed distribution.	100%	Min $0.86, Max $2,171.29
Colour	Text	Colour variant. 10 distinct, 1 unique (Black dominates).	100%	Black, Red, Silver
Subcategory	Text	Product sub-grouping. 37 distinct, 8 unique.	100%	Road Frames, Mountain Bikes
Category	Text	Top-level grouping. 4 distinct: Bikes, Components, Clothing, Accessories.	100%	Components
Background Colour Format	Text	Conditional formatting hex colour for visuals.	100%	#000000
Font Colour Format	Text	Conditional formatting font colour.	100%	#FFFFFF
Category ID	Integer	Numeric ID for category (1–4).	100%	1, 2, 3, 4
Subcategory ID	Integer	Numeric ID for subcategory (1–38).	100%	1 through 38

2.4 Region Table

The Region table maps 10 sales territories to countries and continental groups. It is a lean dimension table with perfect data quality throughout all columns.

Row Count: 10 | Column Count: 4 | Data Quality: 100% Valid, 0% Error, 0% Empty

Column	Data Type	Description	Key Stats	Example
SalesTerritoryKey	Integer	Primary key. 10 distinct, 10 unique. Joins to Sales table.	Range 1–10	1
Region	Text	Sales territory name. 10 distinct, 10 unique.	Min: Australia	Northwest
Country	Text	Country name. 6 distinct, 5 unique. US has multiple regions.	Max: United S…	United States
Group	Text	Continental group. 3 distinct, 1 unique. North America (5), Europe (4), Pacific (1).	3 groups	North America

Countries in Dataset: United States (5 territories: Northwest, Northeast, Central, Southwest, Southeast), Canada, France, Germany, Australia, United Kingdom.

2.5 Sales Table (Central Fact Table)

The Sales table is the core transactional fact table containing 3,616 sales records spanning the full analysis period. It holds all foreign keys linking to dimension tables and the primary numerical columns used in all measures.

Row Count: 3,616 | Column Count: 10 | Data Quality: 100% Valid, 0% Error, 0% Empty across ALL columns

Column	Data Type	Description	Key Stats	Example
SalesOrderNumber	Text	Unique sales order ID. Text format.	3,616 rows	SO-00001
OrderDate	Date	Transaction date. Links to Calendar Table[Date].	Jul 2022–Aug 2025	3/15/2022
ProductKey	Integer	FK to Products table. Links product to sale.	Links 397 SKUs	42
ResellerKey	Integer	Reseller/distributor FK. 103 distinct, 13 unique.	103 distinct	346
EmployeeKey	Integer	FK to Salesperson. 10 distinct, 0 unique.	10 territories	285
SalesTerritoryKey	Integer	FK to Region. 6 distinct, 0 unique.	6 countries	285
Quantity	Integer	Units per transaction. Min 1, Max 21, Avg 2.51, StdDev 1.90.	15 distinct values	2
Unit Price	Decimal	Selling price/unit. Min $5.01, Max $2,146.96, Avg $724.79.	22 distinct, 5 unique	2039.99
Cost	Decimal	Cost/unit. Min $3.40, Max $26,770.16, Avg $1,713.64. High variance.	124 distinct, 37 unique	3824.30
Total Sales	Decimal	Quantity × Unit Price. Pre-calculated revenue column.	Derived column	4079.98

3. DAX Calendar Table — Design & Columns

A dedicated Calendar Table was created using DAX CALENDAR() and ADDCOLUMNS() functions to enable all Time Intelligence calculations. This is a best-practice requirement in Power BI — Time Intelligence functions do not work reliably without a properly marked date table.

3.1 Calendar Table DAX Formula

The Calendar Table was created using the following DAX expression:

```
Calendar Table = ADDCOLUMNS( CALENDAR(MIN(Sales[OrderDate]), MAX(Sales[OrderDate])), "Year", YEAR([Date]), "Month Number", MONTH([Date]), "Month", FORMAT([Date], "MMMM"), "Month Short", FORMAT([Date], "MMM"), "Month Sort", MONTH([Date]), "YearMonth", FORMAT([Date], "YYYY-MM"), "Quarter", "Q" & FORMAT([Date], "Q"), "Year Quarter", FORMAT([Date], "YYYY") & " Q" & FORMAT([Date], "Q") )
```

Configuration: Model View → Select Calendar Table → Table Tools → Mark as Date Table → Select column: Date Relationship: Calendar Table[Date] → Sales[OrderDate] | Type: One-to-Many | Cross-filter: Single

3.2 Calendar Table Column Reference

Column Name	Data Type	Description / Purpose	Example Value	Usage
Date	Date	Primary date column. Foundation for all Time Intelligence. Marked as Date Table column.	7/17/2022	All TI functions
Year	Integer	Calendar year extracted from Date.	2022, 2023, 2024	YoY analysis, slicers
Month Number	Integer	Numeric month 1–12 for correct chart sorting.	1–12	Sort Month column
Month	Text	Full month name. Use Month Sort column for ordering.	January, February	X-axis on seasonal charts
Month Short	Text	Abbreviated 3-letter month for compact visuals.	Jan, Feb, Mar	Compact axis labels
Month Sort	Integer	Duplicate of Month Number. Used explicitly to sort Month column.	1–12	Sort by column setting
Quarter	Text	Quarter label in Q1–Q4 format.	Q1, Q2, Q3, Q4	Quarterly slicers
Year Quarter	Text	Combined year-quarter for unique quarterly filtering.	2023 Q3	Quarter trend analysis

YearMonth	Text	YYYY-MM format for month-level trend charts.	2023-07	X-axis on trend line charts

4. Base Measures & Derived KPIs

Seven foundational measures were created as the building blocks for all Time Intelligence calculations. These measures are used as inputs to all advanced DAX formulas.

Measure Name	DAX Formula	Dashboard Title	Business Purpose
Total Sales	SUM(Sales[Total Sales])	Gross Revenue	Top-line revenue. Foundation for all growth measures.
Total Cost	SUM(Sales[Cost]) OR SUMX(Sales, Sales[Quantity] * Sales[Cost])	Cost of Goods Sold	Direct cost of all units. Input to profit and margin.
Total Quantity	SUM(Sales[Quantity])	Units Sold	Volume metric. Tracks demand independent of pricing.
Total Profit	[Total Sales] - [Total Cost]	Net Profit	Bottom-line after deducting COGS. Key profitability KPI.
Profit Margin	DIVIDE([Total Profit], [Total Sales])	Net Profit Margin %	Percentage of revenue retained as profit.
Average Selling Price	AVERAGE(Sales[Unit Price])	Avg Sell Price (ASP)	Mean selling price per unit. Indicates pricing tier.
Average Cost Per Unit	AVERAGE(Sales[Cost])	Avg Unit Cost	Mean cost per unit sold. Paired with ASP for margin intel.

4.1 KPI Naming Recommendations for Dashboard

Professional business intelligence dashboards use standardised financial terminology aligned with P&L and income statement conventions. Using these titles on your dashboard elevates its credibility for executive and LinkedIn audiences:

Original Label	Recommended Title	Alternative Title	Why This Title
Total Sales	Gross Revenue	Total Revenue Generated	'Gross Revenue' is the standard P&L top-line term. Signals financial sophistication.
Total Profit	Net Profit / Net Income	Operating Income	Globally recognised bottom-line metric. Use 'Net Income' in formal reporting.

Total Cost	Cost of Goods Sold (COGS)	Total Operating Cost	Standard income statement expense line. Positions within P&L framework.

5. Time Intelligence DAX Measures — Full Reference

Sixteen advanced Time Intelligence measures were created to power the four analytical dashboards. Each measure is documented with its full DAX formula, purpose, and business application.

5.1 Year-to-Date (YTD) Measures

Measure	DAX Formula	Business Application
Sales YTD	TOTALYTD([Total Sales], 'Calendar Table'[Date])	Tracks cumulative revenue from Jan 1 to current date. Resets each year.
Profit YTD	TOTALYTD([Total Profit], 'Calendar Table'[Date])	Tracks cumulative profit YTD. Useful for monthly board reporting.
Quantity YTD	TOTALYTD([Total Quantity], 'Calendar Table'[Date])	Tracks cumulative units sold YTD. Useful for inventory planning.
Category Sales YTD	TOTALYTD([Total Sales], 'Calendar Table'[Date]) — with Category in visual	Shows YTD performance per product category. Filters automatically with slicer.
Salesperson YTD	TOTALYTD([Total Sales], 'Calendar Table'[Date]) — with Salesperson in visual	Individual rep's YTD sales total. Used in sales performance dashboard.

5.2 Year-over-Year (YoY) Measures

Measure	DAX Formula	Business Application
Sales PY	CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Calendar Table'[Date]))	Previous year same period sales. Foundation for YoY comparison.
YoY Growth	[Total Sales] - [Sales PY]	Absolute year-on-year revenue growth in currency.
YoY Growth %	DIVIDE([YoY Growth], [Sales PY])	Percentage growth vs prior year. Use for trend line charts.
Country vs Previous Year	CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Calendar Table'[Date])) — with Country	Isolates which countries grew or declined vs prior year.
Region Sales PY	CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Calendar Table'[Date])) — with Region	Regional prior-year comparison for territory management.

5.3 Month-over-Month (MoM) Measures

Measure	DAX Formula	Business Application
Sales PM	CALCULATE([Total Sales], PREVIOUSMONTH('Calendar Table'[Date]))	Previous month sales. Foundation for MoM comparisons.
MoM Growth	[Total Sales] - [Sales PM]	Absolute month-on-month change. Identifies revenue acceleration or deceleration.
MoM Growth %	DIVIDE([MoM Growth], [Sales PM])	Percentage change vs prior month. Use for operational dashboards.

5.4 Rolling & Moving Average Measures

Measure	DAX Formula	Business Application
Rolling 12 Month Sales	CALCULATE([Total Sales], DATESINPERIOD('Calendar Table'[Date], MAX('Calendar Table'[Date]), -12, MONTH))	Trailing 12-month revenue window. Smooths seasonal spikes. Shows long-term growth trajectory.
3 Month Moving Avg	AVERAGEX(DATESINPERIOD('Calendar Table'[Date], MAX('Calendar Table'[Date]), -3, MONTH), [Total Sales])	3-month moving average. Reduces noise. Excellent for trend identification in volatile data.
Best Month	MAXX(VALUES('Calendar Table'[Month]), [Total Sales])	Identifies the single highest-sales month in context. Used in KPI cards and tooltips.

6. Dashboard Analysis — Slide-by-Slide Breakdown

6.1 Slide 2 — Sales Trend by Category & KPI Dashboard

Dashboard Purpose

This dashboard provides the highest-level Time Intelligence overview, displaying 10 KPI cards alongside a multi-category sales trend line chart spanning the full analysis period from July 2022 to August 2025.

KPI Analysis

KPI (Dashboard)	Value	Prof. Title	Interpretation
SUM Total	$77.55M	Gross Revenue	Total revenue across all years. Strong top-line performance.

Total Cost	$76.55M	COGS	$76.55M cost against $77.55M revenue — extremely thin 1.3% margin.
Total Profit	$999.49K	Net Profit	Less than $1M profit on $77.55M revenue — critical margin concern.
Profit Margin	0.01 (1.31%)	Net Profit Margin %	1% margin is unsustainable for most businesses.
Profit Growth	-$240.63K	Profit Growth vs PY	Profits declining vs prior year despite rising revenue.
MoM Growth	$77.55M	Month-on-Month Growth	Current month significantly above prior month.
YoY Growth	$35.13M	Year-on-Year Growth	$35M growth year-on-year — strong revenue trajectory.
Salesperson YTD	$12.65M	Salesperson YTD	Top rep(s) have generated $12.65M YTD cumulatively.
Sales Growth	$35.13M	Absolute Sales Growth	Absolute sales growth aligns with YoY measure.
Sales Growth %	0.83	Sales Growth Rate %	83% revenue growth rate — excellent top-line expansion.

Sales Trend by Category Chart Analysis

The line chart plots YearMonth on the X-axis against Total Sales on the Y-axis, with a separate line for each of the 4 product categories. **Key observations:**

- Bikes: Dominant line showing sustained upward trend from ~$500K (mid 2022) to over $3M (Aug 2025). The steepest growth category, confirming Bikes as the revenue engine.
- Components: Second tier line growing from ~$200K to over $1M by 2025. Consistent growth indicating strong complementary demand alongside Bikes.
- Clothing and Accessories: Near-flat lines close to the x-axis throughout the entire period. Combined contribution barely registers, confirming the extreme category concentration identified in the base report.
- Overall trend direction is bullish — all categories trend upward with no sustained reversals, suggesting healthy market demand.

Strategic Business Decisions — Slide 2

1. Treat Profit Margin as a Crisis KPI: At 1.31%, the business is one supply chain disruption away from losses. Implement an emergency margin review targeting minimum 5% net margin within 12 months.

2. Despite the Profit Growth of -$240.63K, revenue growth at +83% is exceptional. This gap signals costs are rising faster than revenue — conduct a granular COGS analysis by product and region.
3. Leverage the Category slicer to identify if any specific category is disproportionately dragging margin, and implement category-level pricing strategies accordingly.
4. Explore premium pricing strategy for Bikes, where willingness-to-pay may be higher than current pricing reflects.

6.2 Slide 3 — Total Sales & Total Profit by YearMonth

Dashboard Purpose

This page provides granular monthly trend analysis of both revenue (Total Sales) and profitability (Total Profit) using interactive slicers for Month, Year, and Quarter filtering.

Total Sales by YearMonth — Analysis

- Revenue started at ~$0.3M in July 2022 and climbed to a peak of approximately $4.2M in August 2024 — a 14x revenue increase in 26 months.
- Clear upward trend with volatility: sharp month-to-month swings suggest the business operates on non-uniform bulk/seasonal orders rather than predictable recurring revenue.
- The 2023 period shows steady growth from ~$1.5M to $3.5M, with a visible pullback mid-year before recovering.
- 2025 data (partial year) shows $2.0–2.5M per month — slightly below 2024 peaks but above 2022 and 2023 equivalents, indicating sustained growth at a mature level.
- Notable peak: 2024-08 (August 2024) stands as the single highest revenue month at ~$4.2M.

Total Profit by YearMonth — Analysis

- Profit trend is far more volatile than revenue and contains two critical negative-profit periods.
- Most months 2022–2023 show modest positive profits of $50K–$150K, consistent with the thin 1.31% overall margin.
- July 2023: A severe profit crash to approximately -$0.47M. This is the deepest loss period in the dataset and requires urgent historical investigation — likely a high-cost bulk purchase, write-off, or extraordinary expense.
- July–August 2024: A second loss period of -$0.15M to -$0.18M, smaller but still significant. The recurrence in July suggests a seasonal cost pattern — potentially annual supplier cost increases, maintenance cycles, or inventory write-downs.
- 2025 data: Profits have stabilised around $0.02–0.04M per month (thin but positive), suggesting the post-2024 period has normalised.

Strategic Business Decisions — Slide 3

1. Forensic investigation of July 2023 loss (-$0.47M): Pull all transactions from this period and identify the specific cost drivers. Was it a single large-cost order, write-off, or operational expense spike?
2. Establish monthly profit monitoring with automated alerts: Set -$50,000 as an automated Power BI alert threshold on the Total Profit KPI card.
3. Analyse August 2024 peak ($4.2M revenue): Identify which products, regions, and salespeople drove this performance and create a replication playbook.
4. Investigate the recurring July pattern: With losses in both July 2023 and July 2024, there may be a seasonal cost pattern that can be planned for and mitigated in July 2025.

6.3 Slide 4 — Multi-Year Analysis, YoY, Rolling 12M & Seasonality

Total Sales by Month and Year — Clustered Column Chart

This chart displays monthly sales grouped by year, allowing direct year-over-year comparison for each calendar month. Key findings:

- 2024 (dark blue) dominates almost every month — confirming it as the breakthrough year across the entire business.
- August shows the most dramatic year-on-year growth, moving from ~$0.9M (2022) to ~$4.1M (2024).
- 2025 (light blue, partial) is already matching or exceeding 2022 and 2023 performance for equivalent months, suggesting growth momentum is sustained.
- December shows strong multi-year performance consistent with holiday/end-of-year buying patterns.

Year-on-Year Comparison Chart

The clustered column chart plots Total Sales (current year) vs Sales PY (prior year) side by side for each year:

- 2022: $7.8M baseline year — no prior year comparison available.
- 2023: $24.6M current vs $7.8M prior year — a 215% YoY growth rate. Extraordinary expansion.
- 2024: $35.1M current vs $24.6M prior year — a 43% YoY growth rate. Growth rate decelerating but still strong.
- 2025 (partial): $12.6M current vs $35.1M prior — the partial year comparison will appear lower but is pacing appropriately for 5–8 months of data.

Rolling 12-Month Sales Trend

The rolling 12-month line chart is one of the most powerful visuals in the dashboard. It smooths month-to-month volatility and reveals the true long-term revenue trend:

- The rolling line rises continuously from near zero (Jul 2022) to approximately $37M by early 2025 — a perfectly monotonic upward trajectory with no reversals.
- This confirms that despite monthly volatility, the business has never experienced a sustained revenue decline at the 12-month level — a very positive indicator of business health.
- The curve steepens significantly from mid-2023 onwards, corresponding with the YoY growth acceleration identified in the column charts.

Seasonality Analysis

The seasonality line chart plots average monthly sales across all years on the Y-axis with calendar month on the X-axis. Key seasonal patterns:

- Peak months: August ($9.5M cumulative), November ($8.8M), September ($8.2M), and May ($8.3M) — suggesting two seasonal peaks: spring cycling season (April–May) and back-to-school/end-of-summer (August–September).
- Trough months: July ($5.5M), June ($3.5M), and January ($3.8M) — representing the seasonal demand valleys.
- December ($7.5M) shows strong performance, likely driven by holiday gifting and year-end corporate orders.
- This seasonality pattern should directly inform: inventory pre-positioning (stock up before August and November), marketing spend allocation (increase before peak months), and sales team incentive timing.

Strategic Business Decisions — Slide 4

1. Replicate the 2024 playbook: 2024 was a breakthrough year. Conduct a detailed debrief with the sales and operations teams to document what changed — new products, pricing, territories, or salespeople — and institutionalise those practices.
2. Seasonal inventory strategy: Pre-position peak inventory 6–8 weeks before August and November. Negotiate supplier agreements that allow flexible order sizes to match seasonal demand.
3. Holiday campaign planning: December and November consistently perform well. Allocate 25–30% of annual marketing budget to Q4 campaigns to capitalise on seasonal demand.

4. Off-season revenue generation: June and July are consistent troughs. Develop promotional strategies or B2B/corporate partnership sales to generate off-peak revenue.

6.4 Slide 5 — Waterfall Chart: Measure Parameter Analysis

Dashboard Purpose

The Waterfall Chart dashboard is the most advanced analytical page, combining a dynamic measure slicer (10 measures) with a parameter slicer (Product, Category, Country, Region) to enable flexible multi-dimensional analysis. The chart currently displays Profit Margin by Product across years.

Waterfall Chart — Profit Margin by Product (2022–2025)

- 2022 (Total Margin: 0.04): The net margin for 2022 was 4%. Half-Finger Gloves L and S each added +0.31 to margin, as did Men's Bib-Shorts L (+0.31). However, the 'Other' product group collapsed margin by -0.96 — the single largest margin drag in the dataset.
- 2023 (Total Margin: 0.02): Net margin fell to 2%. 'Other' products contributed +0.40 and Classic Vest L added +0.38 of margin uplift. Mountain Bike Socks M (-0.39) and Mountain Bike Socks L (-0.40) together destroyed 0.79 of margin, nearly cancelling all gains.
- 2024 (Total Margin: 0.00): Margin collapsed to effectively zero. Despite 'Other' products contributing a remarkable +1.01 to margin, three specific products (Men's Bib-Shorts L: -0.31, Mountain-Silver 48: -0.32, Classic Vest L: -0.38) collectively destroyed -1.01, perfectly offsetting the gains.
- 2025 (Total Margin: 0.00): Early 2025 data shows zero margin — this continues the 2024 trend.

Key Margin Destroyers Identified

Product	Year	Margin Impact	Action Required
Mountain Bike Socks M	2023	-0.39	Review pricing — selling below cost or near-zero margin
Mountain Bike Socks L	2023	-0.40	Review pricing — consistent margin destroyer
Mountain Bike Socks L	2024	-0.40	Recurring issue — escalate to product/pricing team
Classic Vest L	2024	-0.38	Two-year pattern — consider product discontinuation
Men's Bib-Shorts L	2022 & 2024	+0.31 / -0.31	Mixed signal — investigate cost changes between years

Mountain-Silver 48	2024	-0.32	New margin risk in 2024 — review standard cost
Other Products (Group)	2023-2024	+0.40 to +1.01	Positive contributor — identify and amplify top performers

Strategic Business Decisions — Slide 5

1. Emergency SKU Review — Mountain Bike Socks: Both M and L sizes have been consistent margin destroyers across 2023 and 2024. Conduct a full cost-to-price analysis. If the selling price cannot be raised, consider discontinuing these SKUs or finding a lower-cost supplier.
2. Classic Vest L Pricing Review: Appears as a margin drag in 2024. Calculate actual production cost vs current selling price and either raise the price or discontinue if market price ceiling prevents profitability.
3. Protect the 'Other' Product Group Gains: The Other category contributed +1.01 to margin in 2024. Use the Product slicer to drill down and identify the specific products driving this positive contribution — then scale their sales.
4. Use the Measure Slicer for Board Presentations: Switch to 'Rolling 12 Month Sales' or 'YoY Growth' measures in the waterfall to show the board different profit and growth dimensions in one chart.
5. Establish a Monthly Product Margin Scorecard: Create a Power BI report page showing margin by product for the current month, flagging any product below 5% margin in red for automatic management review.

7. Data Quality Assessment Summary

A comprehensive column quality and profiling analysis was conducted using Power Query's Column Distribution, Column Quality, and Column Profile features across all five tables. The verdict is unambiguous: this dataset is production-quality with zero data issues detected.

Table	Rows	Valid	Error	Empty	Key Profiling Notes
Salesperson	18	100%	0%	0%	EmployeeKey: 18 distinct, 18 unique. Range 272–296. Perfect dimension table.
Date	1,461	100%	0%	0%	MonthNum: 12 distinct months. Even=481, Odd=519. No date gaps detected.
Products	397	100%	0%	0%	Standard Cost: Min $0.86, Max $2,171.29. 131 distinct cost points. 4 categories.
Region	10	100%	0%	0%	10 distinct territories. 6 countries. 3 groups (North America, Europe, Pacific).
Sales [Qty]	3,616	100%	0%	0%	Qty: Min 1, Max 21, Avg 2.51. Right-skewed — most transactions are 1–2 units.
Sales [Unit Price]	3,616	100%	0%	0%	Price: Min $5.01, Max $2,146.96, Avg $724.79. Wide range confirms multi-tier pricing.
Sales [Cost]	3,616	100%	0%	0%	Cost: Min $3.40, Max $26,770.16, Avg $1,713.64. High variance confirms mixed-cost SKUs.

Overall Verdict: No data cleaning or imputation was required. The star-schema design is correctly implemented with proper primary and foreign keys across all tables. The Calendar Table was added as a new dimension to enable Time Intelligence. The only transformation performed was column type casting (ensuring Date is Date type, Integer keys are Integer, Currency columns are Decimal) as confirmed by the Power Query M formula bar visible in the Table Analysis document.

8. Executive Summary & Strategic Recommendations

8.1 Business Performance Assessment

The Bicycle Company generated $77.55M in Gross Revenue across the analysis period (July 2022 – August 2025), representing extraordinary top-line growth from an $7.8M base year (2022) to $35.1M in 2024 alone — a 350% revenue expansion in two years. However, this remarkable revenue achievement is severely undermined by a 1.31% Net Profit Margin, generating only $999.49K in Net Profit on $77.55M of sales.

8.2 Top 5 Strategic Priorities

1. **CRITICAL — Margin Recovery Plan:** With a 1.31% net margin and Profit Growth of -$240.63K, the business is consuming its own growth. Implement a 90-day COGS review targeting 5% net margin.
2. **HIGH PRIORITY — Eliminate Margin-Destroying Products:** Mountain Bike Socks (both sizes) and Classic Vest L have been confirmed margin destroyers across multiple years. Act immediately.
3. **HIGH PRIORITY — Capitalise on Seasonality:** August, November, September, and May are peak months. Pre-position inventory and launch marketing campaigns 6–8 weeks ahead of these peaks.
4. **STRATEGIC — Sustain Revenue Momentum:** 83% sales growth rate is exceptional. Protect the drivers of this growth (top salespeople, top products, top regions) with retention and investment strategies.
5. **PORTFOLIO — Power BI Dashboard Expansion:** Add real-time margin alerts, salesperson leaderboards, and a rolling forecast model to evolve this into a full executive decision-support system.

8.3 Project Documentation Metadata

Field	Details
Project Title	Bicycle Company Sales — Time Intelligence Analysis
Analytics Tool	Microsoft Power BI Desktop
Analytical Method	DAX Time Intelligence — TOTALYTD, SAMEPERIODLASTYEAR, PREVIOUSMONTH, DATESINPERIOD, AVERAGEX
Data Period	July 2022 – August 2025 (37 months)
Total Transactions Analysed	3,616 sales records
Products Covered	397 SKUs across 4 categories
Countries Analysed	6 (United States, Canada, France, UK, Australia, Germany)
Sales Representatives	18 salespeople
DAX Measures Created	7 base measures + 16 Time Intelligence measures = 23 total
Calendar Table Columns	9 (Date, Year, Month, Month Short, Month Number, Month Sort, Quarter, Year Quarter, YearMonth)
Dashboards Built	4 interactive Power BI pages
Document Date	February 2026
Portfolio Platform	LinkedIn Data Analytics Portfolio

This document is part of a professional Data Analytics Portfolio developed by Dr. Abraham Eninla.

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